An ANN-Impedance-based Fault Detection Technique in T-Connection Transmission Overhead lines

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Abstract—**The aim of this paper is to present an impedance-based technique in order to fault detection in transmission overhead lines. When a fault happens in a power system, fast and reliable fault detection methods are not only desirable but also necessary. In this paper, an impedance-based fault detection technique is presented which applied in ANN. In this regard, using the zero sequence impedance values trained by feed-forward back propagation technique, the layer of presented ANN will be trained and after that a fast and accurate fault detection technique is provided. Extended simulations are developed using EMTDC software and the results proved the appropriate performance of the proposed technique in all conditions.**

Keywords- Zero-sequqnce impedance; ANN; fault detection, transmission line.

I. INTRODUCTION

Obviously transmission line is an electrical-mechanical structure that transfers energy in a desired direction. Many different line models are used for a variety of power system analysis applications. The model represents the transmission line impedance depends on type of analysis and type of relay settings [1]. Technically, for setting protective relays and analyzing of protective relays operation, simple frequency dependent impedance is used. The related parameters can be calculated by engineers, or are available in software programs. Moreover, details of shield conductor and ground resistivity are necessary for exact modeling of transmission overhead lines. AC power lines less than 50 miles long are commonly considered electrically short. For the similar cables we can consider them as discrete passive components. Medium-length lines usually are between 50 miles and 150 miles long. Moreover, it can be considered as discrete passive components with a more complex structure by special model. It is obvious that short and medium lines are very important in low-frequency designs such as transmission line modeling while they are modeled with lumped-circuit theories. As mentioned before, transmission lines are classified into short, medium and long lines for the purpose of modeling. In definition, electrically long line is a line which is electrically long if its physical length is comparable to the wavelength transmission line equations [2]. Also long line can derive a set of circuit equations for a differential slice of the mentioned model and also differential equations can be developed to describe the voltage and current in the transmission line. If the line transposed, for transposed sections, the total voltage drop for each conductor is the same thus eliminating negative- and zero-sequence voltages and currents, so this case will be more complex [3]. So far, researches show that double circuit lines create more challenges because of the effects of mutual inductance which is not entirely eliminated by the transposition. Generally, transmission lines have four electrical parameters as resistance, inductance, capacitance and conductance. The effects of magnetic and electric fields around the conductor lead to some variations on inductance and capacitance. The shunt conductance characterizes the leakage current through insulators, which is very small and can be neglected. The parameters R, L and C are necessary not only during planning for the development of the transmission line modeling due to power system analysis, but also operation stages. While the resistance of the conductor is best determined from manufactures data, the inductances and capacitances can be evaluated using related formula. In this paper, the transmission lines are reproduced by an approximate equivalent model with exact parameters on per phase in the EMTDC software. Using these model, voltages, currents, power flows, efficiency and regulations can be computed.

Nowadays Neural Network analysis and applications are extended widely and many documents are published in this field [4-5]. In fact the applications of Neural Network are increasing in power system to bring artificial intelligence in power systems. Also, many formulas are presented in order to calculate zero sequence impedance of power components such as power transformer and transmission line. In [6], the principle of directional earth fault protection using zero sequence transients in non-solid earthed network is presented and in [7], faulty line selection by comparing the amplitudes of transient zero sequence current in the special frequency band for power distribution networks is analyzed. Also [8] studies principle of directional earth fault protection using zero sequence transients in non-solid earthed network. Moreover, [9] studies faulty line selection by comparing the amplitudes of transient zero sequence current in the special

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frequency band in power distribution networks. Ref [10] introduces line impedance measurement device and some of related methods. Moreover in [11], a mutual inductance based technique is presented to electrical parameter estimation. All of the methods are either complex or not practically simple; so researches for overhead line fault detection are still open.

II. ZERO SEQUENCE IMPEDANCE ESTIMATION

Technically, two measurement values are sufficient to calculate positive and zero sequence impedances in power systems. Similar measurements on each phase can be performed to obtain better estimations of self and mutual impedances by averaging method which presented as following:

$$Z_S(f)= (Z_{AA}(f) + Z_{BB}(f) + Z_{CC}(f))/3 \tag{1}$$

$$Z_M(f)=(Z_{AB}(f)+Z_{AC}(f)+Z_{BA}(f)+Z_{BC}(f)+Z_{CA}(f)+Z_{CB}(f))/6 \tag{2}$$

where:

Z_{AA} = Self impedance of the conductor
Z_{AB} = Mutual impedance between conductor A and B
Z_{AC} = Mutual impedance between conductor A and C
$Z_S(f)$= Average self impedance
$Z_M(f)$= Average mutual impedance

Values of self and mutual impedances can be estimated and averaged, so positive and zero sequence impedances of a symmetrical or transposed line can be calculated as equations (3) and (4):

$$Z_1(f)= Z_S(f) – Z_m(f) \tag{3}$$

$$Z_0(f)= Z_S(f) + 2 Z_m(f) \tag{4}$$

where:

$Z_1(f)$= Positive sequence impedance
$Z_0(f)$= Zero sequence impedance

In case of un-transposed or non-symmetrical lines, the measurement provide full 3x3 matrix of line impedances, which can be converted into sequence component matrix using symmetrical component transformations. In case of long lines or cable, impedance can also be obtained with similar method. After extended and complicated calculations, zero impedance of double circuit lines without wire grounding, with one wire and with two wires are calculated as following formulas, but because of the limitation of space, we cannot show all the formulas and only the final term of calculation of zero impedance is presented. For double circuit lines without grounding wire we have equation (5) as following:

$$Z_{0(ag)} = R_0 + JX_0 = \frac{R'}{n_2} + \frac{3}{4}\pi f \mu_0 (3Ln\frac{\delta}{\sqrt[3]{r_B D_M^2}} + \frac{\mu T}{4n_2}) + \frac{3\pi f\mu}{f} + j f Ln\frac{\delta}{\sqrt{D_{M1}D_{M2}}})) \tag{5}$$

where in equations (5) to (10), the used parameters have following definitions:

$D_{M1} = \sqrt[3]{D_{Aa} D_{Bb} D_{Cc}}$, $D_{M2} = \sqrt[3]{D_{Ab} D_{Ac} D_{Bc}}$, f= frequency, μ = magnetic field constant, μ_E = magnetic field constant of air, μ_0 = magnetic field constant of conductor, δ = return depth of earth current, r_B =bundled radius, r_E = conductor resistance, T= conductor temperature and finally n_2 = conductor number. For double circuit with one grounding wire, equation (6) can be written as following:

$$Z_{0(ag)} = R_0 + JX_0 = \frac{R'}{n_2} + \frac{3}{4}\pi f \mu_0 (3Ln\frac{\delta}{\sqrt[3]{r_B D_M^2}} + \frac{\mu T}{4n_2}) + \frac{3\pi f\mu}{f} + j f Ln\frac{\delta}{\sqrt{D_{M1}D_{M2}}})) - 6\frac{(\frac{3\pi f\mu}{f} + j f Ln\frac{\delta}{\sqrt{D_{M1}D_{M2}}}))^2}{R'_E + \frac{\mu_E}{4} + j f Ln\frac{\delta}{r_E})} \tag{6}$$

For double-circuit with two grounding wires, equation (7) can be written as following:

$$Z_{0(ag)} = R_0 + JX_0 = \frac{R'}{n_2} + \frac{3}{4}\pi f \mu_0 (3Ln\frac{\delta}{\sqrt[3]{r_B D_M^2}} + \frac{\mu T}{4n_2}) + \frac{3\pi f\mu}{f} + j f Ln\frac{\delta}{\sqrt{D_{M1}D_{M2}}})) - 6\frac{(\frac{\pi f\mu}{4} + j f Ln\frac{\delta}{D_{M2E}}))^2}{\frac{R'_E}{2} + \frac{\pi f\mu}{4} + j f \left[Ln\frac{\delta}{r_E} + \frac{\mu_E}{8}\right])} \tag{7}$$

For four-circuit with two grounding wires, equation (8) can be written as following:

$$Z_0 = Z_{0(a)} - \frac{Z^2{}_{0(ag)}}{Z^2{}_{0(g)}} \tag{8}$$

where in equation (8), the related terms will be described as equations (9) and (10) as following:

$$Z_{0(g)} = \frac{r_e}{2} + 0.00477 \quad f + j0.01397 \quad f \ Log_{10}\frac{D_e}{\sqrt{(GMR)_{Conductor} \ (GMD)_{Separation}}} \tag{9}$$

$$Z_{0(ag)} = 0.00477 \quad f + j0.01397 \quad f \ Log_{10}\frac{D_e}{d_{ab}} \tag{10}$$

As above, the zero-sequence impedance of a transmission line having ground wires is calculated. The calculated values are compared with the measured zero-sequence impedance of a line and the results are satisfactory. One of the results is that the actual zero-sequence impedance is different from the zero-sequence impedances of the lines considering either no ground wire or continuous ground wire. This difference is significant if the ground wire is of ACSR conductor.

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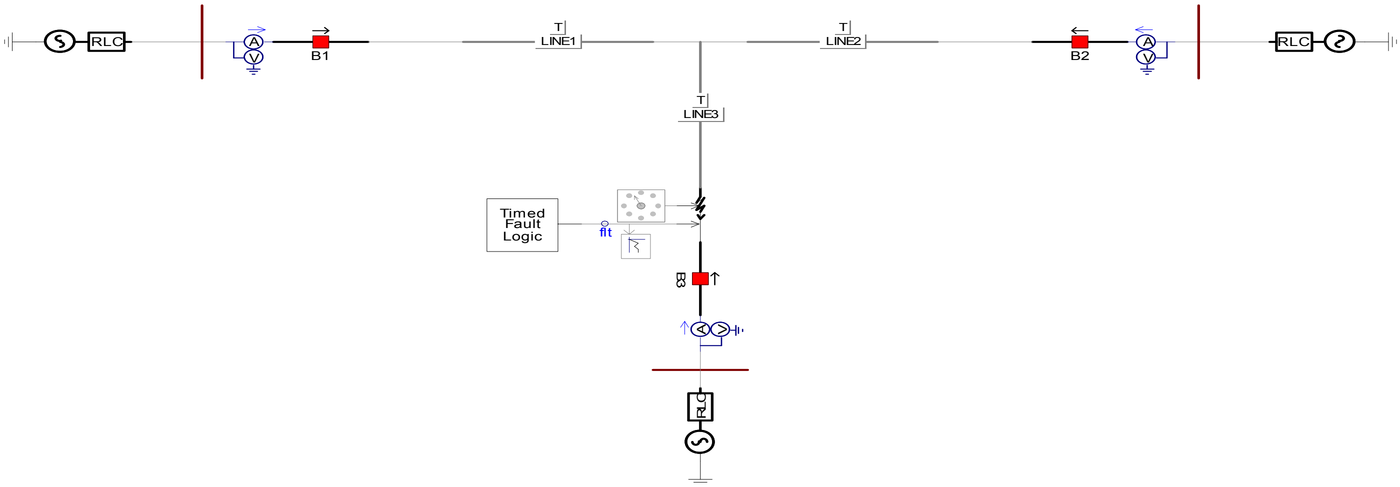

Fig. 1. Simulated power system schematic.

III. ANN ARCHITECTURE

In this work, a feed-forward ANN specifically has been designed for fault detection of transmission overhead lines. To investigate the performance of designed method, a network has been trained and simulated in PSCAD/EMTDC and the related codes are written in MATLAB software. After an off-line training process using a back propagation technique, the network was connected to an electrical system for on-line detection of transmission overhead faults. The results of a will verify the excellent performance of the designed network.

A) ANN Learning

To get the intended outputs for the given inputs, the network weights need to be adjusted. The process of weights adjustment or called network training iteratively is performed by presenting a collection of input data and desired output data. This type of learning is called supervised learning. The weight update can be done in either batch or incremental mode. In epoch mode, the weights are updated after all learning data in the training set which has been presented. The incremental mode updates the weights every time a data in the collection is presented. Two issues in updating weights are as following; the first one is when to stop updating, and the second one is when to stop the training, and how the weights are changed.

The training can be stopped in two ways that the first one is using maximum epoch and the second one is using a cost function. An epoch refers to a complete training data collection. Training data for 20 epochs means the weights are updated with the learning rule continuously until input data set has been presented for 20 times. A cost function is a performance measurement and network learning commonly uses the Mean Square Error method.

B) Backpropagation

Back propagation network (BPN) is a feed-forward network training using back propagation technique. The back propagation algorithm developed by Clelland and Rummelhart in 1986 used gradient descent learning rule to calculate and update the weights. In this method, during training, each input is forwarded due the intermediate layer until outputs are resulted. Each output is then compared to the desired output to get the errors that will be transmitted backwards to the intermediate layer that contributes directly to the output. Based on these errors, the weights are calculated and updated. This method is repeated layer by layer until each node in the network reached a minimum error signal that defines its relative contribution to the total error. Using back propagation technique, weights adjustment is resulted. The gradient descent algorithm suffers from slow training time. Some other useful and fast algorithms such Levenberg-Marquardt, Quasi-Newton, and conjugate gradients algorithms have been used to optimize the learning rules in BPN while the network training is done using already developed MATLAB software.

IV. RESSULTS

As shown in Fig. 2, the rms value of Z_1, from now on referred as Z, will be the input for the neural network. A data window of 20 samples will be the network input. The voltage and current are sampled from current transformer (CT) and potential transformer (PT) for 20 samples per cycle. The data window consists of 20 samples or 1 waveform. The input layer of the ANN relay is fed from the 20 sample data window, which forms a 2 cycles Z curve pattern. The next input data pattern is obtained by shifting the data window by one sample. For each fault case, 40 data patterns covering 20 pre-fault and 20 fault data were generated for training and testing the network. As a result, it should be noted that after getting the V_{rms} and I_{rms} , the frequencies are doubled which double the frequency of Z waveform as well. Therefore, using 20 sample data window for Z curve will cover 2 cycles of the curve. However, this is of no concern from this point of view for pattern recognition. The output file from PSCAD software contains Z values, which changes with respect to time step of 1 ms. The windowing process is done using MATLAB program. The output layer is intended to output 1 (trip) or 0 (no-trip).

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Table 1: THE PARAMETERS OF T-CONNECTION LINE [7]

Case	Power system branches	
Impedance	Resistance (ohm)	Reactance (H)
Branch 1	30	R= 0.4, L=4mH
Branch 2	40	R=0.4, L=4mH
Branch 3	50	R=0.4, L=4 mH

Table 2: MEASUREMENT RESULTS

Case	Power system branches		
branch	Branch 1	Branch 2	Branch 3
Setting Value	12+j37	16+j50	20+j62
Measuring Value	12.4+j35.9	16.7+j50.4	20.2+j62.5
Error	1%	1%	1%

Table 3: MEASUREMENT RESULTS WITH TRADITIONAL METHOD

Case	Power system branches		
branch	Branch 1	Branch 2	Branch 3
Setting Value	12+j37	16+j50	20+j62
Measuring Value	13+j39	16.9+j58.3	19.3+j60.2
Error	5%	4%	6%

TABLE 4: RELAY OPERATION

Case	Power system branches		
branch	Branch 1	Branch 2	Branch 3
1	Trip	NO	NO
2	NO	Trip	NO
3	NO	NO	Trip

The output layer only has one node with log-sigmoid transfer function, which is also used in every node in all the layers. To generate binary output, a hard limit function with a 0.5 threshold is attached to the network output when the network is in use, i.e. hard limit function is not used during training. In network training, the (input, desired output) pair of data is fed to the input layer and output layer. The choice of transfer function determines the value ranges and the desired output can reach. For instance, using hard limit transfer function where the output is 0 or 1, the idealistic output cannot have value of 0 or 1 because this will make the error to be zero all the time. Using either 0.99 is a better choice of desired output; however, it gives error value of either 0.01 or 0.99. As opposed to the hard limit function, the log-sigmoid is a continuous function, which maps the input to a value ranged from 0 to 1. The error generated by the difference between the desired output and the output can vary between 0 to 1. The high degree of variety in error is necessary in back-propagation training. The log-sigmoid transfer function is the most suitable to use in ANN relay application where the desired output can only have 0 or 1 value.

From Fig. 2, the similarity of the waveforms between various fault types can be seen. The curves that have similarities are grouped as (BG and CG), (AB, BC, and AC), (ABG, ACG, and BCG), and (ABC and ABCG). Based on the mentioned discussion, the fault types included in the training data were AG, BG, AB, AC, ABG, and ABC. A part from the fault cases, a normal case was included in the training data as well. Thus, the number of training data patterns is 16,000 consisting of 15,500 fault data patterns and 500 normal data patterns. The network input layer

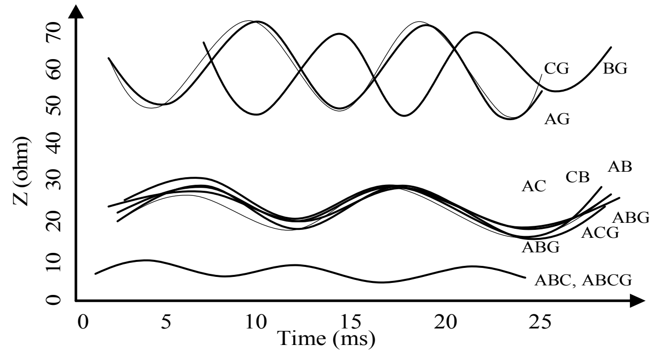

Fig. 2. Waveforms of various fault types.

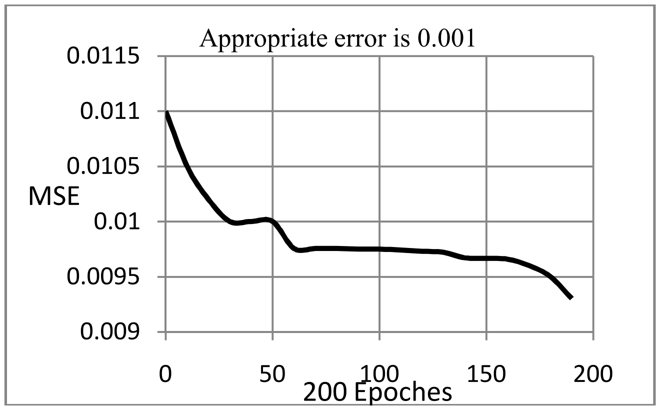

Fig. 3. Learning curve of network.

requires pair of input data. The desired output data set was generated by setting a pickup value below which a trip (1) signal is generated. Fig. 2 shows the Z waveforms for all types of fault and distance. Further reducing the hidden layer size to five nodes did not improve the learning rate and the performance. The network of this size cannot map the input from the training data to desired output. Also Fig. 3 shows the learning curve of network. The training did not reach the given approach, the training was stopped at 200 epochs with MSE=0.068. At this stage, testing the network using the trained and untrained data sets

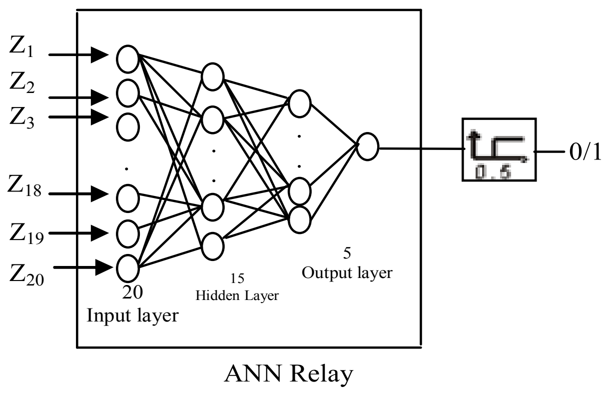

Fig. 4. ANN relay architecture.

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TABLE. 5: NETWORK SIMULATION SUMMARY

Network Size	Final MSE	Epochs	Time (min)	Testing Error			
				Using Trained Data Set (26,620)		Using Untrained Data Set (17,600)	
				No	%	No	%
[20 20 10 1]	0.07	403	31	N/A	N/A	N/A	N/A
[20 20 10 1]	0.006	1000	128	25	0.1	28	0.2
[20 15 5 1]	0.01	17	4	52	0.2	44	0.3
[20 15 5 1]	0.002	167	19	5	0.02	20	0.1
[20 15 1]	0.002	35	6	17	0.1	22	0.1
[20 5 1]	0.07	200	10	2585	9.2	1651	9.4

gave 2,875 and 1,257 wrong outputs. Since the network cannot be reduced further without suffering the performance and the training rate, no more new network size was trained. The network simulation resulted in the ANN relay architecture with an input layer of 20 nodes, 2 hidden layers of 15 and 5 nodes and an output layer of 1 node as shown in Fig. 4. The hard limit function is connected to the network output to ensure 0/1 output. The weights of the network are different from the previous. The graphs show the weight distribution in each node in a layer. In the hidden layer graphs, x-axis represents the node in the layer. The weight distribution of the first node in the layer is shown in node 1 of the x-axis. Also, all the results of the training and testing are given here for convenience. The network with sizes of [20 20 10 1] and [20 5 1] are either too big or too small. Both of them cannot map the training data to the intended output satisfactorily. It can be seen from the learning curves that they hardly converged anymore when the training was stopped. The smaller network hardly converged anymore when it reached 0.07 MSE, thus the performance of the network suffered greatly with about 7.8 % output error when tested using simulation data set. The bigger network performed better with average of 0.12 % error, however the training time was long. The other two network architectures [20 15 5 1] and [20 15 1] had better performance and faster convergence.

V. CONCLUSION

The applications of Neutral Network for calculation of zero sequence impedances have been demonstrated in this paper in overhead transmission lines with and without grounding wires. This type of work has complex nature, but using the described technique, appropriate results can be achieved. The simulation results show that the values of zero sequence impedances trained by ANN can be used for locating the faults especially in T-Connection transmission overhead lines with grounding wires.

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APPENDIX

TABLE 1. THE POWER SOURCE PARAMETERS

	Src1	Src2	Src3
Source impedance type	R	R	R
Base MVA (3-phase)	100 MVA	100 MVA	100 MVA
Base voltage L-L (rms)	230 KV	230 KV	230 KV
Base frequency	50 Hz	50 Hz	50 Hz
Zero sequence included	No	No	No
Impedance data format	RRL Values	RRL Values	RRL Values

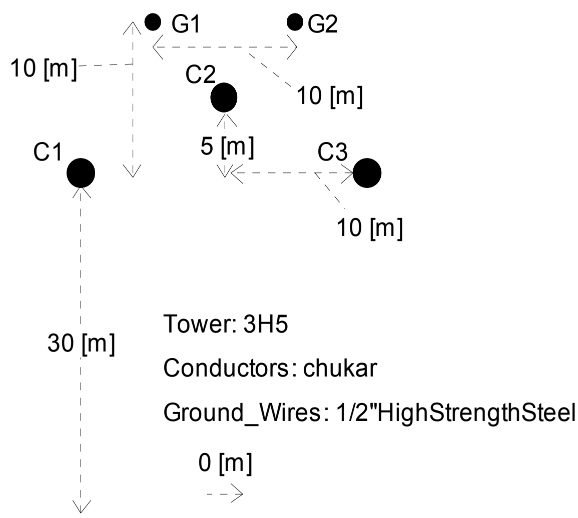

Fig. 1. Conductor orientation.

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TABLE 3. TL GROUND WIRE DATA

Ground Wire No	1	2
Conductor Name	7/16 Steel	7/16 Steel
Cond Radius(cm)	0.55245	0.55245
Horiz. Dist. X(m)	-5	5
Height at Tower Y(m)	35	35
Sag at Midspan(m)	10	10
DC Resistance(ohms/km)	2.8645	2.8645

TABLE 4. TL CONDUCTOR DATA

Bundle No	1	2	3
Conductor Name	Chukar	Chukar	Chukar
Conductor Type (AC/DC)	AC	AC	AC
V(kV)(AC:L-L,rms/DC:L-G,pk)	500	500	500
V Phase (Deg)	0	-120	120
Line I (kA)(AC:rms/DC:pk)	5	5	5
Line I Phase (Deg)	20	-100	140
No of Sub-Conductors	2	2	2
Sub-Cond Radius (cm)	2.03454	2.03454	2.03454
Sub-Cond Spacing (cm)	45.72	45.72	45.72
Horiz. Dist. X(m)	-10	0	10
Height at Tower Y(m)	30	30	30
Sag at Midspan(m)	10	10	10
DC Resistance(ohms/km)	0.03206	0.03206	0.03206

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